Sri Lanka Institute of Information Technology

Trends in Digital Media

Assignment - 01

SE4051 – Trends in Digital Media

Registration Number	Name
IT22152732	Hearth H.M.H.S

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Title of the Environment

Adaptive Smart Study Room: An Intelligent and Human-Centered Interactive Environment

Scenario / Application Domain

The proposed system represents a smart study environment designed to improve student productivity and focus. This environment simulates a digital study room where the system continuously monitors user behavior and adapts the environment accordingly. The application is particularly useful for students engaged in self-study or remote learning, where maintaining focus and discipline is often challenging. By integrating intelligent interaction and adaptive responses, the system transforms a static study space into an interactive and supportive learning environment.

Target User

The primary target users of this system are university students and individuals engaged in self-learning activities. These users often face challenges such as distractions, lack of motivation, and inefficient study habits. The system is designed to assist such users by providing real-time feedback, guidance, and environmental adjustments to improve their focus and productivity.

Intelligent Behavior

The system demonstrates intelligent behavior by dynamically monitoring and responding to user interactions within the environment. It detects when the user enters or leaves specific areas such as the study zone, break zone, and completion zone. A focus score is dynamically calculated based on user behavior, including time spent studying and periods of inactivity. The system evaluates this score to determine the user’s level of engagement and provides appropriate feedback. Additionally, the system generates context-aware suggestions and warnings, ensuring that the user is guided effectively throughout the study session.

Adaptive Features

The environment adapts dynamically based on user interaction. Several adaptive features are implemented, including lighting changes, audio feedback, and visual indicators. For example, the lighting becomes brighter and more focused during study mode, while it becomes softer during break periods. Audio cues are used to reinforce different states such as study, break, and completion. A visual study indicator highlights the active study area, guiding the user intuitively. Furthermore, the system provides random smart suggestions and alerts based on user performance, creating a responsive and engaging environment.

Intended User Experience

The intended user experience is designed to be intuitive, supportive, and motivating. The user is guided from the beginning through clear instructions and visual cues. As the user interacts with the environment, they receive real-time feedback that helps them maintain focus and improve their study habits. The system avoids overwhelming the user by providing clear and meaningful messages rather than excessive notifications. Overall, the environment aims to create a sense of awareness and engagement, helping users stay disciplined and productive in a natural and user-friendly manner.